Seq@K: Why Pass@K Can Pick the Wrong Model for an Agent That Revises

Language-model agents routinely operate in a loop where the model proposes an answer, a user or verifier points out what failed, and the model revises (Section 1). Standard leaderboard metrics such as Pass@1 and Pass@K (Chen et al., 2021) cannot see that loop. For each task prompt x_i in a benchmark of N tasks paired with binary verifier V_i, Pass@K restricts the model context strictly to x_i, draws K independent candidate answers from the actor distribution _(. | x_i), and scores 1 if max_{1 t K} V_i(Y_{i,t}) = 1 (Section 3, Equation 1). Every attempt starts from the same static prompt with no record of earlier attempts or why they failed. Ranking models for iterative workflows by Pass@K therefore assumes that one-turn accuracy predicts how well a model uses feedback (Section 1).

On 27 October 2026, Thomson Yen, Gilbert Yang, Naimeng Ye, Minghui Chen, Dong Xia, Tianyi Shen, Yechan Jeon, Yaqin Chen, Daksh Mittal, Fangyu Wu, and Hongseok Namkoong at Columbia Business School, the University of California, Santa Barbara, and Northwestern University submitted "Seq@K: Evaluating Language Models Under Sequential Feedback" (arXiv:2610.37859v1). Long-horizon agent benchmarks such as SWE-bench, WebArena, -bench, and MINT score planning, tool use, state tracking, and feedback handling together (Section 2). Seq@K isolates iterative revision of whole attempts under explicit feedback across 7 benchmark variants, 16 frontier models, and 471 completed runs (Abstract, Section 2, Section 4, Table 1).

How Seq@K is defined and the three feedback modes

Section 3 separates the actor model being evaluated from any critic model that writes feedback and any verifier V_i that grades attempts. Under Seq@K, a feedback mode m specifies what message F_{i,t} the environment returns after a failed attempt (V_i(Y_{i,t}) = 0). Before attempt t, the prompt expands with the interaction history H_{i,t-1} = ((Y_{i,1}, F_{i,1}), ..., (Y_{i,t-1}, F_{i,t-1})), starting from empty history H_{i,0} on attempt 1. Attempts t 2 are revisions sampled from _(. | x_i, H_{i,t-1}). When an attempt passes verification, the trajectory stops early. Seq@K computes the expected trajectory success rate max_{1 t K} V_i(Y_{i,t}) over the trajectory distribution induced by the actor, feedback mode m, and attempt budget K (Section 3, Equation 2).

Because each revision conditions on prior context, Seq@K forms a family of feedback-conditioned evaluations indexed by m rather than a single standalone score. A numerical Seq@K value is well-defined only after specifying the feedback mode m alongside the benchmark task split, verifier, prompt builder, decoding hyperparameters, attempt budget, and history retention policy (Section 3, Section 5.3.2). Across instruction following (AdvancedIF), rubric-graded reasoning, medical, and research tasks (CLBench DKR, CLBench RSA, HealthBench, ResearchRubrics), and self-invoking Python synthesis (MBPP Pro, HumanEval Pro), the main experiments run at decoding temperature T = 0.7 and compare Pass@K against three sequential feedback modes (Section 4, Appendix A.1):

- **Binary feedback.** The actor receives only a pass or fail verdict after a failed attempt, such as a fixed line stating that its Python code failed the hidden tests (Section 4, Appendix A.2). This mode tests whether knowing an attempt failed suffices to guide revision.
- **Raw verifier feedback.** The environment returns direct evaluator diagnostics, including per-item rubric verdicts on AdvancedIF or up to two failed test calls with

expected outputs, actual outputs, or error messages truncated at 300 characters in coding (Section 4, Appendix A.1, Appendix A.2).

- **LLM critic feedback.** Either the actor itself or another language model writes targeted critique as directive instructions or socratic probing questions, with explicit access to the grading rubric (rubric-aware) or without it (rubric-blind) (Section 4, Section 5.2.1, Appendix A.1).

Model selection, performance gaps, and retry correlations (Tables 2 and 3)

Model rankings shift once actors can revise from feedback (Section 5.1.1, Figure 1). On 30 shared `AdvancedIF` tasks with raw verifier feedback, `Sonnet 4.6` rises from last place among 7 actors under `Pass@5` (`0.500` in Table 11) to the top tied group under `Seq@5` (`1.000`), with rank inversions also occurring across 6 actors on 28 shared `CLBench DKR` tasks (Figure 1, Table 10). To test whether sequential evaluation picks better models for sequential deployment, Section 5.1.1 simulates calibration across 4 coding actors (`Sonnet 4.6`, `DeepSeek V3.2`, `Gemini 3 Flash`, and `Qwen3 Coder Flash`) repeated 3 times each at `T = 0.7` across 48 runs and 2,040 trajectories on 45 `MBPP Pro` tasks (split into groups of 22 and 23) and 40 `HumanEval Pro` tasks (split into two groups of 20) (Appendix B.1). Both protocols stop charging API costs at the first passing attempt. With `n = 10` calibration tasks on `MBPP Pro` (Table 2, Table 8), `Pass@5` calibration averages $0.407 and selects an actor reaching 90.2% held-out test-detail `Seq@5`, whereas `Seq@3` calibration averages $0.383 and reaches 98.4% (+8.3 percentage points, 95% bootstrap interval `[+1.8, +19.9]`, with `Seq@4` reaching 98.2% at $0.432, +8.1 points). On `MBPP Pro`, calibrating with `Seq@3` on 1 task also matches the selection loss of `Pass@5` on 10 tasks at 11x lower cost (Figure 2). On `HumanEval Pro` at `n = 10` (Table 2, Table 8), `Pass@5` calibration costs $1.015 for 70.8% held-out `Seq@5`, while `Seq@3` costs $0.805 for 96.8% (+26.0 points, `[+6.8, +44.7]`) and `Seq@4` costs $0.937 for 96.7% (+25.9 points).

`Pass@5` can either overstate or understate differences between actors (Section 5.1.2, Table 3, Appendix B.2, Table 9). On `AdvancedIF` with raw verifier feedback, the gap between the best and worst actor shrinks from 33.3 points under `Pass@5` to 6.7 points under `Seq@5` (-26.7 points, 95% interval `[-43.3, -3.3]`), and the average all-pairs gap changes by -11.7 points (`[-21.0, -1.6]`). Coding with failed-test details moves the other way. On `MBPP Pro`, the best-to-worst gap widens from 18.5 to 31.1 points (+12.6, `[-7.4, +29.6]`, all-pairs +8.6), and on `HumanEval Pro` it expands from 10.8 to 51.7 points (+40.8, `[+12.5, +52.5]`, all-pairs +25.4). Cost-accuracy Pareto frontiers for `K = 1` through `5` also move upward under sequential feedback (Section 5.1.3, Figure 3). On the 28 shared `CLBench DKR` tasks, independent sampling solves at most about 14% of tasks (`0.143` for `DeepSeek V4 Pro` and `GPT-5.2` in Table 10), whereas `Seq@5` reaches 50% (`0.500` for `Opus 4.7`).

Section 5.1.4 isolates why the two protocols diverge. Across 26 non-coding model-benchmark pairs at `K = 5` and `T = 0.7`, raw `Pass@5` and `Seq@5` have a Spearman correlation of `+0.68` (Pearson `+0.81`, Figure 4a, Appendix C.2) because both numbers include attempt-1 accuracy. Subtracting the first attempt to compare multi-turn gains from attempts 2 through 5, `Pass@5 = Pass@5 - Pass@1` versus `Seq@5 = Seq@5 - Seq@1`, drops the Spearman correlation to `-0.08` (95% interval `[-0.47, +0.31]`, Pearson `-0.04`, Figure 4b, Appendix C.2). Eleven of the 26 pairs record zero gain after attempt 1 under `Pass@5` (Appendix C.2, Table 11). In the paper's words, how much a model gains from more independent samples says little about how much it gains from feedback.

When revisions gain and how they fail

Models also differ in when their sequential gains arrive (Section 5.2.1). For configurations with `Seq@5 > 0`, Early-Gain Share `EGS = (Seq@2 - Seq@1) / Seq@5` measures the fraction of

improvement earned on the first retry, and Last-Gain Share LGS = (Seq@5 - Seq@4) / Seq@5 measures the fraction earned on attempt 5. On 30 CLBench DKR tasks (Table 4), Gemini 3 Flash with socratic GPT-5.2 critique earns most of its progress immediately, posting 0.86 EGS and 0.00 LGS as accuracy moves from 0.033 at Seq@1 to 0.267 at Seq@5. Opus 4.7 under directive self-critique climbs from 0.000 to 0.500 with 0.33 EGS and 0.07 LGS, and DeepSeek V4 Pro under socratic GPT-5.2 critique climbs from 0.067 to 0.400 with 0.30 EGS and 0.10 LGS. Appendix D.1 shows that EGS and LGS belong to the actor-feedback pair rather than the actor alone, since Qwen3.6 Max records 0.20 EGS in Table 4 and 0.57 and 0.50 under two other critic and prompt setups.

Some revisions fail without changing a single character (Section 5.2.2). Across three runs each on HumanEval Pro and MBPP Pro, Qwen3 Coder Flash and DeepSeek V3.2 resubmit the exact previous failed code unmodified on 48% and 6% of revised attempts respectively, whereas Sonnet 4.6 and Gemini 3 Flash never submit an unmodified revised attempt (Figure 5a). Other revisions edit the response while leaving the criticized defect untouched. On an AdvancedIF task requiring a side-hustle assistant to decline off-topic questions, GPT-5.2 answers an off-topic question about AI and painting on all 5 attempts and swaps only the recommended side hustle, whereas Gemini 3 Flash adds the refusal on attempt 3 and passes on attempt 4 (Section 5.2.2).

To quantify revision quality across benchmarks, the authors develop a labeling prompt from 14 audited examples and 11 random development tasks and use Claude Opus 5.5 to annotate revisions across 102 held-out tasks from AdvancedIF, CLBench DKR, HumanEval Pro, and MBPP Pro (Section 5.2.2, Appendix D.2, Appendix E.1). Across 842 valid annotated revisions, 186 (22%) are unhelpful and satisfy none of the valid feedback requests, 289 (34%) address some of the feedback, and 367 (44%) address all of the feedback (Section 5.2.2, Appendix D.2). Many revisions that address every request in the feedback still fail on multi-criterion benchmarks such as CLBench DKR and AdvancedIF, because the edit breaks a criterion that passed on the previous turn (Figure 5b). In an 18-requirement CLBench DKR podcast compliance run (Appendix E.3, Example E.2), GPT-5.2 reduces its failed requirements from 4 items to 1 over 5 attempts ({4, 8, 11, 16} to {11}), Attempt 5 fixes the group-generalization rule R_4, but it proposes rewrites where the grader expects escalation, so rule R_11 (refrain from fixing content with serious ethical concerns) fails.

Diagnosing the feedback loop in Tables 5 and 6

Seq@K also diagnoses whether the feedback channel gives the actor information it cannot derive alone (Section 5.3). On CLBench DKR, Section 5.3.1 measures criterion repair as the share of requirements failed on attempt 1 that pass in the actor's best later attempt, pooled across 5 models with 30 tasks each (Table 5, Appendix F.1, Table 15). Independent sampling repairs 36.5% of failed criteria, binary feedback repairs 34.4%, and a rubric-blind critic repairs 36.8%. A rubric-aware critic informed by the grading criteria repairs 73.8%, while criterion regression stays between 1.8% and 2.9% across all four conditions (Table 5, Table 15). Whether binary pass or fail feedback helps depends on both the benchmark and the actor (Section 5.3.1, Appendix F.2, Table 16). Across six models on CLBench DKR, binary feedback raises Seq@5 over Pass@5 by only 1 point on average, and in coding it shifts Seq@5 by at most 3.3 points for DeepSeek V3.2 and Qwen3 Coder Flash. For Sonnet 4.6, binary feedback raises Seq@5 over Pass@5 by 17 to 26 points on MBPP Pro (0.637 to 0.807) and HumanEval Pro (0.358 to 0.617), as the model reasons on its own about why its code failed once told tests did not pass (Section 5.3.1, Table 16).

Summarizing earlier attempts instead of keeping the full transcript in context can reduce multi-turn accuracy (Section 5.3.2). Across 57 matched run pairs yielding 2,520 trajectory

pairs, replacing full previous attempts with a summary changes success rates very little on AdvancedIF (-0.5 percentage points across 200 pairs), MBPP Pro (+1.8 points across 800 pairs), and HumanEval Pro (+0.5 points across 800 pairs) (Table 6, Appendix F.3, Table 17). On rubric-heavy domain and medical benchmarks, summarization lowers sequential success by 6.1 percentage points on CLBench DKR (180 pairs), 7.2 points on ResearchRubrics (180 pairs), and 14.4 points on HealthBench (360 pairs) (Table 6). How interaction history is retained is therefore part of the evaluation protocol and needs to be reported alongside Seq@K (Section 5.3.2).

Scope limits and practitioner takeaways

The paper itself documents the limits of these findings. First, arXiv:2610.37859v1 is a 27 October 2026 preprint evaluated on specific benchmarks and small task subsets: 30 shared tasks on AdvancedIF, 28 or 30 tasks per actor on CLBench DKR (20 for GPT-5.5 Pro and 29 for Qwen3.6 27B), 10 tasks on CLBench RSA, 30 or 60 tasks on HealthBench, and 30 tasks on ResearchRubrics (Table 10, Table 11). Second, the coding analyses cover 45 MBPP Pro and 40 HumanEval Pro tasks after excluding 15 of 100 tasks where tests call function names absent from or mismatched with the prompt. Appendix A.3 notes that other task defects can remain, and the history-summarization comparison in Appendix F.3 uses all 50 tasks per coding benchmark at T = 0 rather than T = 0.7. Third, the Table 2 selection experiment covers only the 4 coding actors against Pass@5 rather than every way of spending an independent-sampling budget. In that test, Seq@3 and Seq@4 are read from K = 5 runs whose prompts stated five attempts were available, and Seq@4 was added after viewing Seq@3 results. Three of the four cost-ratio 95% intervals include one, with Seq@3 costing more than Pass@5 on 28% of drawn MBPP Pro calibration sets (Appendix B.1, Table 8), while the MBPP Pro gap-change intervals in Table 9 include zero.

Fourth, the revision annotations and non-coding graders come from language models rather than humans. Claude Opus 5.5 produced the 842 revision labels, and Claude Sonnet re-labeled a 15% stratified sample (165 revisions) with 80% agreement (= 0.73, falling to = 0.53 on CLBench DKR), while three evaluated actors are Claude models and requests the annotator did not tie to a rubric item may slightly overstate the broken-criteria counts (Appendix D.2). Non-coding verifiers likewise rely on Sonnet 4.6 for AdvancedIF and GPT-5.4 with a GPT-5.2 fallback for CLBench (Appendix A.1). Fifth, the 26-pair correlation analysis excludes coding, picks each pair's highest-scoring non-binary feedback run on the same tasks (making Seq@5 values optimistic, Appendix C.2), and resamples the 26 pairs as independent despite shared actors and benchmarks. Several CLBench DKR comparisons also pair different actors with different critic configurations and exclude critic API charges from DKR frontier costs (Appendix C.1, Appendix D.1), while raw verifier and rubric-aware modes supply privileged grading information rather than testing unguided self-correction (Section 2, Appendix A.1).

The paper's recommendation for practitioners building iterative agent workflows follows directly from these protocol dependencies (Section 6). Teams should select models with Seq@K under the feedback signal and history policy the deployed agent will receive, because ranking candidates by Pass@K can favor a model that scores well on single attempts yet uses feedback poorly (Section 5.1.1, Section 6). When designing the feedback loop, practitioners benefit from highlighting evaluation criteria that the model would not derive on its own from reasoning over a bare failure verdict (Section 5.3.1). Because history summarization can lower sequential accuracy on rubric-heavy benchmarks, every reported Seq@K score should specify its feedback mode, history retention, task split, verifier, decoding hyperparameters, and attempt budget (Section 3, Section 5.3.2).

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